Time to Event Models

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R
```

```r
conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

In many health sciences applications, binary[1] outcomes[2] are *incompletely observed*. For example, if we are studying whether cancer patients experience a relapse after an initial remission, we may not be able to follow patients to the end of their lives; instead, we may only know whether each patient has relapsed before the end of the study. If a patient has not relapsed by that point, we might not know if they will relapse at some other date or if they will stay cancer-free for the rest of their lives. [3] Their recurrence status at end-of-life is *missing data*. If some study participants withdraw from a study before the scheduled end date, additional missing data arise. Standard logistic regression[4] cannot handle such right-censored[5] data directly without introducing bias[6].

However, these outcome observations are not *entirely* missing. We know that those patients stayed relapse-free *at least* until the time point when they were last observed. If we also know the *time-to-event* for participants who experienced events while under study, we can analyze the *time-to-event-or-study-exit* alongside an event indicator using *survival analysis*. The survival analysis framework provides the foundation for the topics covered in this module.

1 Why censoring matters

Example 1.1 (Six patients followed for relapse)**.** Suppose a study follows six cancer patients in remission for up to 12 months, and we want to estimate the 12-month relapse risk, $1 - S(12)$, where S is the survival function[a] of time to relapse (in months). Table 1 shows each patient's observed follow-up time and event indicator (1 = relapse observed; 0 = right-censored[b]).

Table 1: Hypothetical follow-up data for six patients

Patient	Follow-up time (months)	Event indicator	Relapse status at 12 months
A	3	1	relapsed

[a] intro-to-survival-analysis.qmd#def-surv-fn
[b] intro-to-survival-analysis.qmd#def-right-censor

[1] https://morrison-lab.github.io/sds/data.html#def-binary-var
[2] Intro-to-GLMs.qmd#def-outcome
[3] Binary outcomes are typically defined *for a specific time-point*. It is important to clearly define whether we are interested in outcome status at end of study, at end of life, or at some other time.
[4] logistic-regression.qmd#def-logistic-regression
[5] intro-to-survival-analysis.qmd#def-right-censor
[6] https://morrison-lab.github.io/sds/estimation.html#def-bias

B	5	0	unknown (withdrew)
C	7	1	relapsed
D	9	0	unknown (withdrew)
E	12	0	relapse-free
F	12	0	relapse-free

Naive approach 1: treat censored patients as event-free. Counting patients B and D as relapse-free gives $2/6 \approx 0.333$. This approach can only *under*estimate the risk, because any relapses that B or D experienced after withdrawal are counted as non-relapses.
Naive approach 2: drop patients with unknown status. Discarding patients B and D gives $2/4 = 0.5$. This approach throws away the information that B stayed relapse-free for 5 months and D for 9 months, so in this example it *over*estimates the risk.
Using the censored observations. The Kaplan-Meier estimator[c] keeps each censored patient in the risk set[d] until their censoring time [1, Ch. 2]. Relapses occur at 3 months (6 patients at risk) and at 7 months (4 patients at risk: C, D, E, F), so

$$\begin{aligned}
\hat{\mathrm{S}}_{\mathrm{KM}}(12) &= \left(1 - \frac{1}{6}\right) \cdot \left(1 - \frac{1}{4}\right) && \text{(product over the event times 3 and 7)} \\
&= \frac{5}{6} \cdot \frac{3}{4} && \text{(subtract inside each factor)} \\
&= \frac{15}{24} && \text{(multiply numerators and denominators)} \\
&= 0.625 && \text{(divide)}
\end{aligned}$$

and the estimated 12-month risk is $1 - 0.625 = 0.375$. That estimate lies between the two naive estimates, and it is valid under non-informative censoring[e] [1, Ch. 1].

[c] intro-to-survival-analysis.qmd#def-KM-estimator
[d] intro-to-survival-analysis.qmd#def-risk-set
[e] intro-to-survival-analysis.qmd#def-noninformative-censoring

2 Module roadmap

This module has three chapters.

1. Introduction to Survival Analysis[7] defines the basic quantities for time-to-event data: right censoring[8], the survival function[9], the hazard function[10], and the cumulative hazard[11]. It then covers nonparametric methods: the Kaplan-Meier estimator[12], the log-rank test[13], and the Nelson-Aalen estimator[14].
2. Proportional Hazards Models[15] introduces the Cox proportional hazards model[16], which relates covariates[17] to a baseline hazard[18] through hazard ratios[19], and estimates its coefficients using the partial likelihood[20]. It also introduces competing risks[21].
3. Parametric Survival Models[22] assumes a specific distributional family for the event times,

[7] intro-to-survival-analysis.qmd
[8] intro-to-survival-analysis.qmd#def-right-censor
[9] intro-to-survival-analysis.qmd#def-surv-fn
[10] intro-to-survival-analysis.qmd#def-hazard
[11] intro-to-survival-analysis.qmd#def-cuhaz
[12] intro-to-survival-analysis.qmd#def-KM-estimator
[13] intro-to-survival-analysis.qmd#def-logrank-test
[14] intro-to-survival-analysis.qmd#def-na-cuhaz-est
[15] proportional-hazards-models.qmd
[16] proportional-hazards-models.qmd#def-ph-model
[17] Intro-to-GLMs.qmd#def-predictors
[18] proportional-hazards-models.qmd#def-base-hazard
[19] intro-to-survival-analysis.qmd#def-hazard-ratio
[20] proportional-hazards-models.qmd#def-ph-partial-lik
[21] proportional-hazards-models.qmd#def-competing-risks
[22] parametric-survival-models.qmd

such as the exponential regression model[23] and the Weibull distribution[24], and introduces accelerated failure time models[25].

[1] D. G. Kleinbaum and M. Klein, *Survival analysis: A self-learning text*, 3rd ed. Springer, 2012. Available: https://link.springer.com/book/10.1007/978-1-4419-6646-9

[23] parametric-survival-models.qmd#def-exp-reg
[24] https://morrison-lab.github.io/pds/distributions.html#def-weibull
[25] parametric-survival-models.qmd#def-aft